V3 empirical study · Instruction

Five models · four underlyings · four volatility regimes · 1.5-year fixed calibration · return-based calibration

What this report is

- Companion to the seven-model 10,000-path V4 fixed-calibration study. This report keeps GBM, Modified GBM, GARCH, Merton, and GARCH–Merton, whose P dynamics are estimated from lookback returns. That is the same V3 split: return-based vs option-implied Heston, not a split by the P→Q map. GBM still uses $\mu \to r_f$; GARCH uses Duan LRNVR; Merton uses Pan jump-size premium μ_J^* from listed calls on top of return-based P jumps; GARCH–Merton is unchanged from V3.
- Companies are SPY, AAPL, MSFT, and AMZN. No new calibration or LSM.
- Option-implied companion: V4_1p5y_fixed_empirical_study_option_implied.pdf.
- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 5 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

Experimental design (held fixed for every model)

- Models: GBM, Modified GBM (Markov direction), GARCH(1,1), Merton jump-diffusion, GARCH–Merton.
- Modified GBM (Markov direction + split-normal magnitudes) is specified in V3_modified_gbm_model.pdf in this folder.
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at the first session of each 1-year evaluation window. Rolling: none (one estimate, held for the whole window). Parameters estimated at t0 are used only after t0.
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start

V3 empirical study · Instruction (continued)

Five models · four underlyings · four volatility regimes · 1.5-year fixed calibration · return-based calibration

Shared market sample and filters (not model-specific)

- For each company × regime, one listed-call sample is drawn once and reused by all five models in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S, K, C; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq DTE \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05, valid bid–ask with relative spread ≤ 50% when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: (κ, θ, ξ, ρ, v_0) are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \to r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud (n_paths = 10000, seed 42, n_steps = DTE, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{model} - C_{mkt}) / C_{mkt})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE} = \text{mean}\,|C_{model} - C_{mkt}|$ (dollars). $\text{Bias \$} = \text{mean}(C_{model} - C_{mkt})$ (dollars). $\text{Bias\%} = 100 \times \text{mean}((C_{model} - C_{mkt})/C_{mkt})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all five models in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-3	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-3	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-3	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-3	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7–60	|S/K−1|≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end − 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with trading_date ≤ t. LSM prices the Monday contract using the latest calibration row with date ≤ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid–ask mid when the source mark is $0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All five models in this report still share that exact sample.

Heston / Heston–Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity × moneyness strata). GBM / Merton / GARCH / GARCH–Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	39.85	1.5118	-1.3570	-29.27	25.5%	3
Modified GBM	43.13	1.6255	-1.5635	-35.30	26.1%	5
GARCH	36.15	1.3736	-0.9849	-18.26	23.5%	2
Merton	40.12	1.5229	-1.3790	-29.99	24.9%	4
GARCH–Merton	**35.69**	**1.3398**	**-0.8186**	**-13.54**	**20.8%**	**1**

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	40.10	0.7145	+0.5358	+27.51	26.2%	2
Modified GBM	41.67	0.7464	+0.5698	+29.03	26.1%	4
GARCH	**38.23**	**0.6786**	**+0.4831**	**+25.30**	**24.6%**	**1**
Merton	40.53	0.7231	+0.5464	+27.95	24.2%	3
GARCH–Merton	45.88	0.8274	+0.6986	+34.18	22.4%	5

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	23.62	0.8840	-0.1277	+1.81	20.8%	4
Modified GBM	23.03	0.8902	-0.2283	-0.59	22.2%	3
GARCH	25.60	1.0389	-0.8655	-16.06	18.9%	5
Merton	**22.81**	**0.9091**	**-0.4865**	**-6.72**	**17.7%**	**1**
GARCH–Merton	22.86	0.9032	-0.4350	-5.93	16.7%	2

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	57.52	2.9208	-0.5111	+18.64	24.3%	2
Modified GBM	59.99	2.9847	-0.3345	+21.88	23.9%	3
GARCH	60.68	2.9733	-0.3077	+22.51	22.9%	4
Merton	**56.81**	**2.9399**	**-0.5830**	**+17.28**	**20.3%**	**1**
GARCH–Merton	70.25	3.1595	+0.2869	+33.51	20.3%	5

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths e

Page 5 · Table block · Company summary · five models × four regimes

Each number is the arithmetic mean of AAPL, MSFT, AMZN. Same columns as the SPY page. Bold green row = lowest mean RMSE%. 1.5-year lookback, rolling=none, n_paths=100

Table Co-1 Company summary · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · mean of AAPL, MSFT, AMZN · n̄ = 50

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	32.08	1.0927	-0.1692	-4.59	23.9%	4
Modified GBM	30.51	1.0614	-0.4437	-9.68	25.3%	2
GARCH	31.21	1.0899	-0.0949	+2.20	22.4%	3
Merton	**30.09**	**1.0744**	**-0.2531**	**-4.56**	**18.7%**	**1**
GARCH–Merton	43.19	1.3542	+0.6781	+21.13	17.4%	5

Table Co-2 Company summary · 2013-2014 Normal
2014-01-01 → 2014-12-31 · mean of AAPL, MSFT, AMZN · n̄ = 50

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	47.76	2.2220	+1.4315	+36.60	24.6%	4
Modified GBM	44.74	2.0480	+1.2472	+33.61	25.7%	2
GARCH	**42.14**	**1.7764**	**+0.9287**	**+30.56**	**23.3%**	**1**
Merton	45.10	2.2098	+1.3747	+33.84	18.7%	3
GARCH–Merton	66.53	2.7630	+2.5061	+57.71	17.5%	5

Table Co-3 Company summary · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · mean of AAPL, MSFT, AMZN · n̄ = 51

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	20.48	4.6263	-2.0225	-2.87	23.3%	3
Modified GBM	**20.18**	**4.6410**	**-1.3699**	**-3.86**	**23.8%**	**1**
GARCH	20.19	4.6326	-2.0914	-5.32	21.8%	2
Merton	22.32	4.7292	-2.3122	+2.55	13.8%	4
GARCH–Merton	23.79	5.1311	+1.3847	+8.36	16.6%	5

Table Co-4 Company summary · 2019-2020 COVID
2019-09-01 → 2020-08-31 · mean of AAPL, MSFT, AMZN · n̄ = 50

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	48.08	10.4481	+4.7656	+23.77	24.2%	2
Modified GBM	48.43	10.8481	+5.6520	+23.33	24.2%	3
GARCH	**41.87**	**9.2660**	**-1.0332**	**+15.36**	**22.4%**	**1**
Merton	49.62	10.6941	+5.4435	+25.70	19.8%	4
GARCH–Merton	55.47	10.1345	+2.6265	+32.21	19.4%	5

RMSE%, MAE, Bias $, Bias%, and early-exercise fraction are each averaged across AAPL, MSFT, AMZN. This is not a pooled re-pricing of a combined contract sample. Ranking uses the mean RMS

Page 6 · Table block · AAPL · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	34.70	1.7325	-0.2012	+2.49	23.4%	3
Modified GBM	**32.72**	**1.6706**	**-0.6554**	**-4.84**	**24.4%**	**1**
GARCH	35.04	1.7531	-0.1784	+2.83	21.8%	4
Merton	34.57	1.7293	-0.2406	+1.71	21.0%	2
GARCH–Merton	43.53	2.0289	+0.7015	+16.92	16.7%	5

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	57.70	3.8194	+3.8194	+51.38	24.4%	4
Modified GBM	51.10	3.3484	+3.3484	+44.85	25.7%	2
GARCH	**39.62**	**2.4884**	**+2.3946**	**+32.33**	**23.1%**	**1**
Merton	57.50	3.8718	+3.8718	+51.52	19.1%	3
GARCH–Merton	67.13	4.6054	+4.6054	+61.25	17.2%	5

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**16.77**	**0.8809**	**-0.5584**	**-7.11**	**23.8%**	**1**
Modified GBM	16.88	0.8919	-0.5478	-6.91	24.5%	2
GARCH	17.11	0.9076	-0.6227	-8.15	22.6%	3
Merton	18.08	0.8489	+0.2167	+5.33	14.1%	5
GARCH–Merton	17.34	0.8370	+0.0909	+3.31	19.3%	4

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**41.39**	**3.1111**	**+0.6446**	**+18.83**	**24.0%**	**1**
Modified GBM	43.50	3.2399	+0.9173	+21.54	24.2%	2
GARCH	45.91	3.3764	+1.2410	+24.56	22.0%	4
Merton	43.70	3.2565	+0.9786	+22.02	17.8%	3
GARCH–Merton	68.46	5.0306	+3.8961	+50.08	18.7%	5

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths e

Page 7 · Table block · MSFT · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	29.17	0.3120	-0.2729	-19.88	25.5%	4
Modified GBM	28.96	0.3119	-0.2681	-19.42	26.3%	3
GARCH	26.24	0.2848	-0.0629	+0.36	23.1%	2
Merton	**26.16**	**0.2932**	**-0.1837**	**-11.78**	**18.4%**	**1**
GARCH–Merton	32.88	0.2963	+0.0849	+14.04	20.0%	5

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	62.49	0.4284	+0.4262	+52.88	24.7%	3
Modified GBM	60.58	0.4138	+0.4100	+51.07	26.0%	2
GARCH	64.23	0.4441	+0.4431	+54.76	23.9%	4
Merton	**55.50**	**0.3746**	**+0.3666**	**+45.96**	**17.5%**	**1**
GARCH–Merton	96.85	0.7307	+0.7307	+87.77	17.1%	5

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	21.68	0.5626	+0.0129	+3.25	23.0%	3
Modified GBM	**19.94**	**0.5524**	**-0.2155**	**-3.65**	**25.4%**	**1**
GARCH	20.42	0.5496	-0.1917	-2.97	21.3%	2
Merton	25.35	0.6315	+0.2531	+10.52	12.3%	4
GARCH–Merton	25.36	0.6383	+0.2795	+11.22	16.3%	5

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	45.98	1.7242	+0.0264	+16.93	24.8%	3
Modified GBM	**41.04**	**1.7043**	**-0.3996**	**+8.84**	**25.5%**	**1**
GARCH	42.17	1.6816	-0.2778	+11.08	22.6%	2
Merton	46.60	1.7641	+0.0420	+17.39	21.5%	4
GARCH–Merton	53.39	1.8833	+0.4719	+25.84	18.4%	5

RMSE% is $100\times\sqrt{\text{mean}(((\text{model}-\text{market})/\text{market})^2)}$. MAE and Bias $ are in option-premium dollars. Bias% is $100\times\text{mean}((\text{model}-\text{market})/\text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 8 · Table block · AMZN · five models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, rolling=none, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	32.38	1.2336	-0.0334	+3.64	22.9%	4
Modified GBM	29.85	1.2017	-0.4075	-4.77	25.2%	2
GARCH	32.34	1.2318	-0.0435	+3.42	22.3%	3
Merton	**29.53**	**1.2007**	**-0.3350**	**-3.61**	**16.7%**	**1**
GARCH–Merton	53.16	1.7374	+1.2478	+32.42	15.6%	5

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	23.09	2.4183	+0.0489	+5.54	24.8%	4
Modified GBM	22.52	2.3818	-0.0168	+4.90	25.5%	2
GARCH	22.58	2.3967	-0.0514	+4.59	22.8%	3
Merton	**22.30**	**2.3829**	**-0.1145**	**+4.05**	**19.7%**	**1**
GARCH–Merton	35.61	2.9530	+2.1822	+24.10	18.2%	5

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	**22.97**	**12.4353**	**-5.5218**	**-4.77**	**22.9%**	**1**
Modified GBM	23.72	12.4787	-3.3465	-1.02	21.5%	4
GARCH	23.04	12.4406	-5.4598	-4.84	21.5%	2
Merton	23.53	12.7074	-7.4064	-8.19	14.9%	3
GARCH–Merton	28.66	13.9181	+3.7835	+10.56	14.3%	5

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias $	Bias%	Early-ex. frac.	Mark
GBM	56.86	26.5091	+13.6258	+35.54	23.7%	3
Modified GBM	60.74	27.6000	+16.4384	+39.62	22.9%	5
GARCH	**37.53**	**22.7400**	**-4.0628**	**+10.43**	**22.5%**	**1**
Merton	58.57	27.0617	+15.3098	+37.68	20.1%	4
GARCH–Merton	44.57	23.4896	+3.5116	+20.71	21.1%	2

RMSE% is 100×√mean(((model−market)/market)²). MAE and Bias $ are in option-premium dollars. Bias% is 100×mean((model−market)/market). Early-ex. frac. is the mean LSM share of paths e

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells bes
1	**GARCH**	**35.43**	**35.59**	**3**
2	Modified GBM	37.46	36.88	3
3	Merton	37.60	37.34	6
4	GBM	37.89	37.28	3
5	GARCH–Merton	46.35	44.05	1

Table S2 · Wins by company (lowest RMSE% in that company×regi

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
GBM	0	2	0	1	3
Modified GBM	0	1	2	0	3
GARCH	1	1	0	1	3
Merton	**2**	**0**	**2**	**2**	**6**
GARCH–Mertor	1	0	0	0	1

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	34.03	45.85	21.26	50.44	37.89
Modified GBM	33.66	43.97	20.89	51.32	37.46
GARCH	**32.44**	**41.17**	**21.54**	**46.57**	**35.43**
Merton	32.59	43.96	22.44	51.42	37.60
GARCH–Merton	41.32	61.37	23.55	59.17	46.35

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Page 8 · Summary by company · mean RMSE% across four regimes

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
GBM	40.27	39.98	0
Modified GBM	41.95	42.40	0
GARCH	40.17	37.19	1
Merton	**40.07**	**40.32**	**2**
GARCH–Merton	43.67	40.78	1

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
GBM	37.64	38.04	2
Modified GBM	36.05	38.11	1
GARCH	**34.42**	**37.33**	**1**
Merton	38.46	39.13	0
GARCH–Merton	49.12	55.33	0

Table S4.3 · MSFT

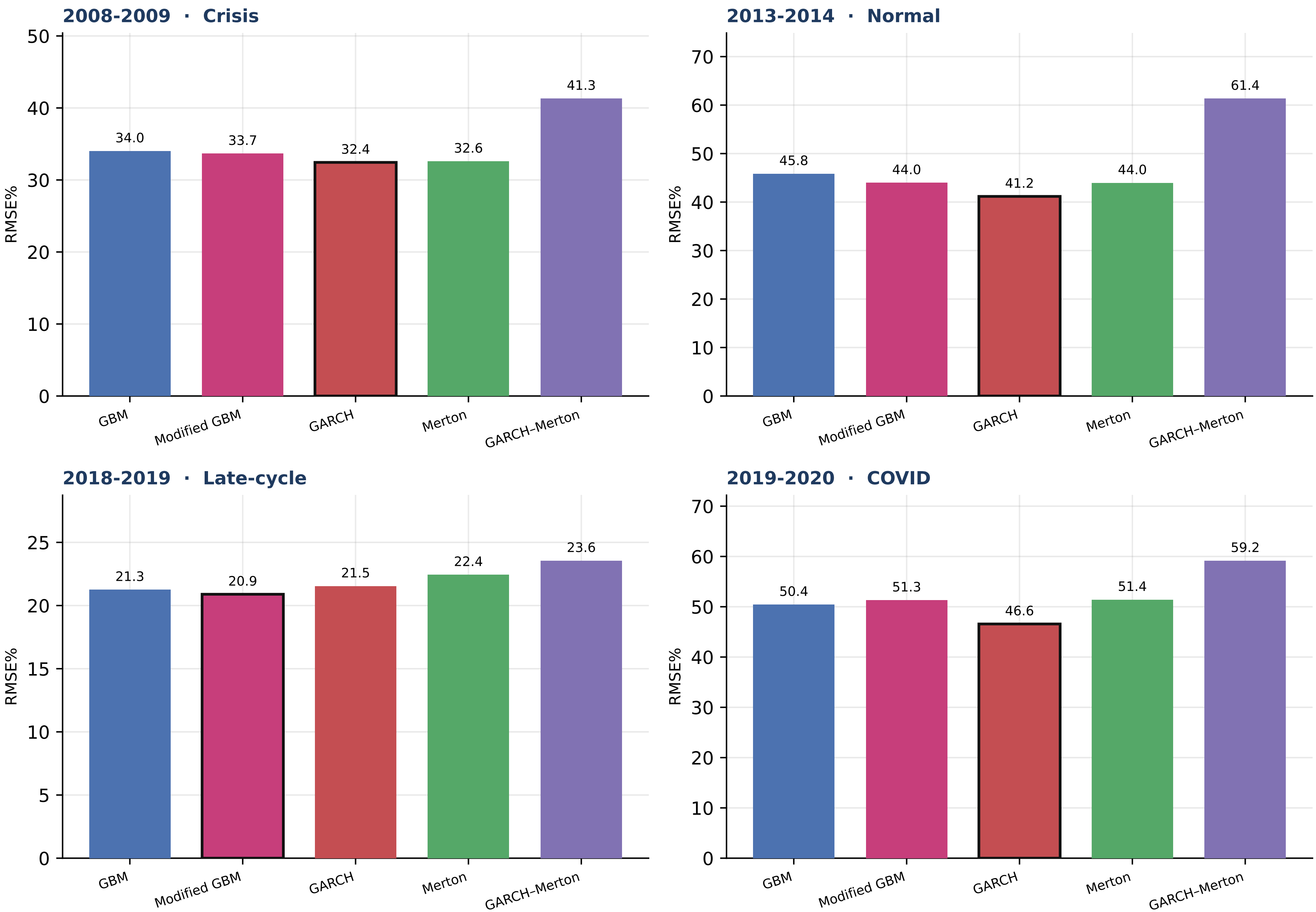
Model	Mean RMSE%	Median	# regimes best
GBM	39.83	37.57	0
Modified GBM	**37.63**	**35.00**	**2**
GARCH	38.27	34.21	0
Merton	38.40	36.38	2
GARCH–Merton	52.12	43.14	0

Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
GBM	33.82	27.73	1
Modified GBM	34.21	26.79	0
GARCH	**28.87**	**27.69**	**1**
Merton	33.48	26.53	2
GARCH–Merton	40.50	40.09	0

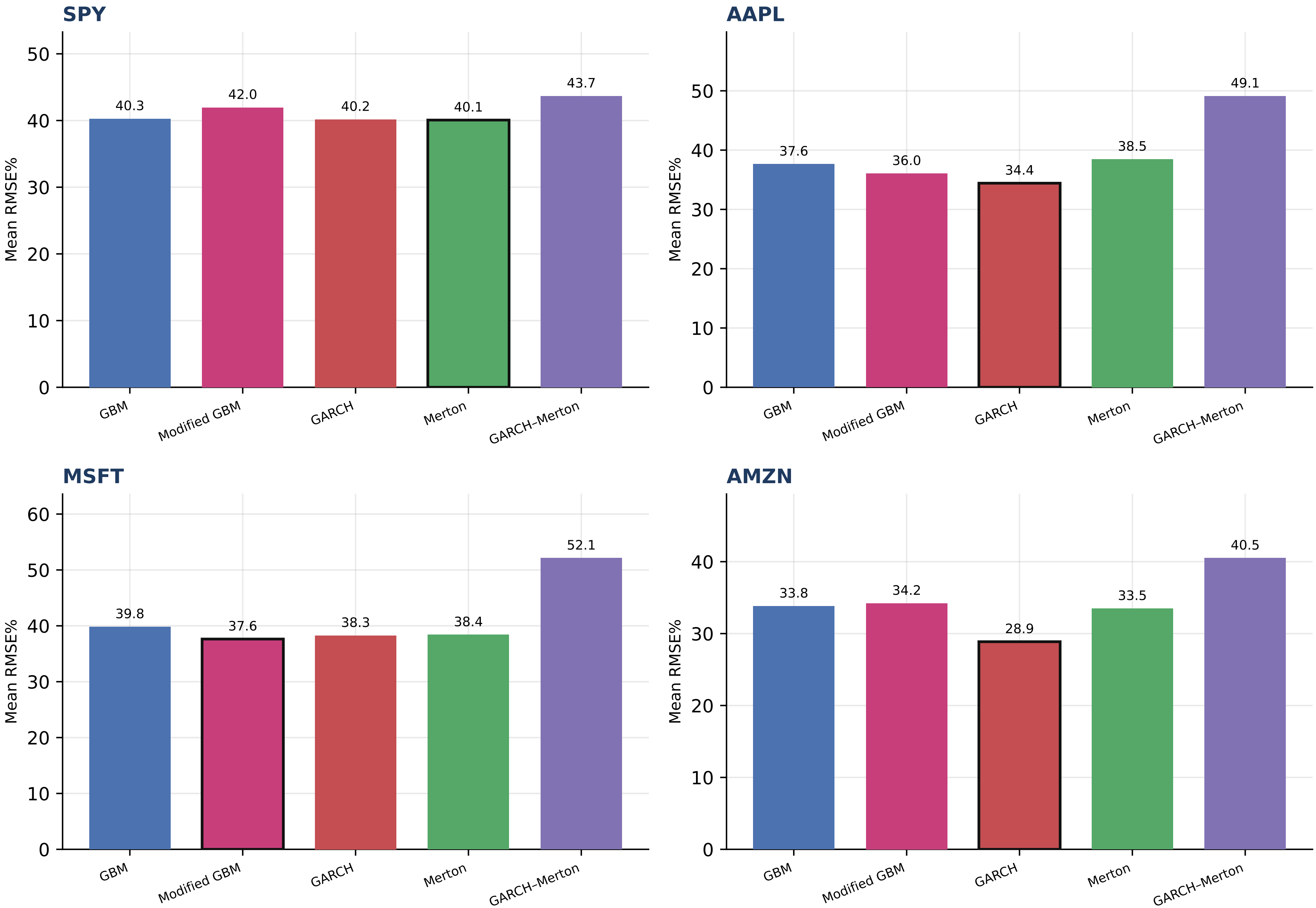
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.

Page 9 · Figure 1 · RMSE% by model and regime · mean of SPY / AAPL / MSFT / AMZN



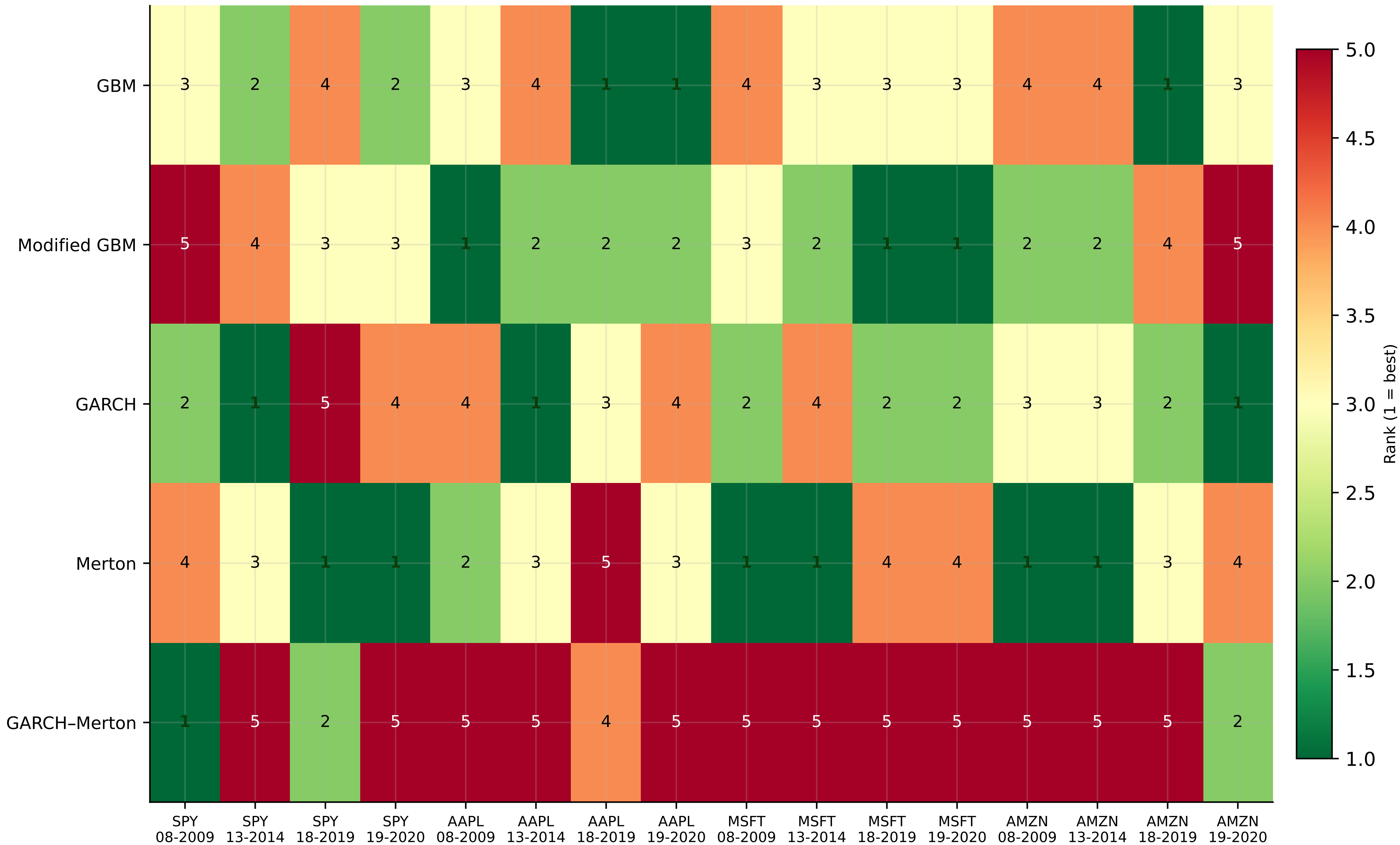
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.

Page 10 · Figure 2 · RMSE% by model · each company (mean across four regimes)



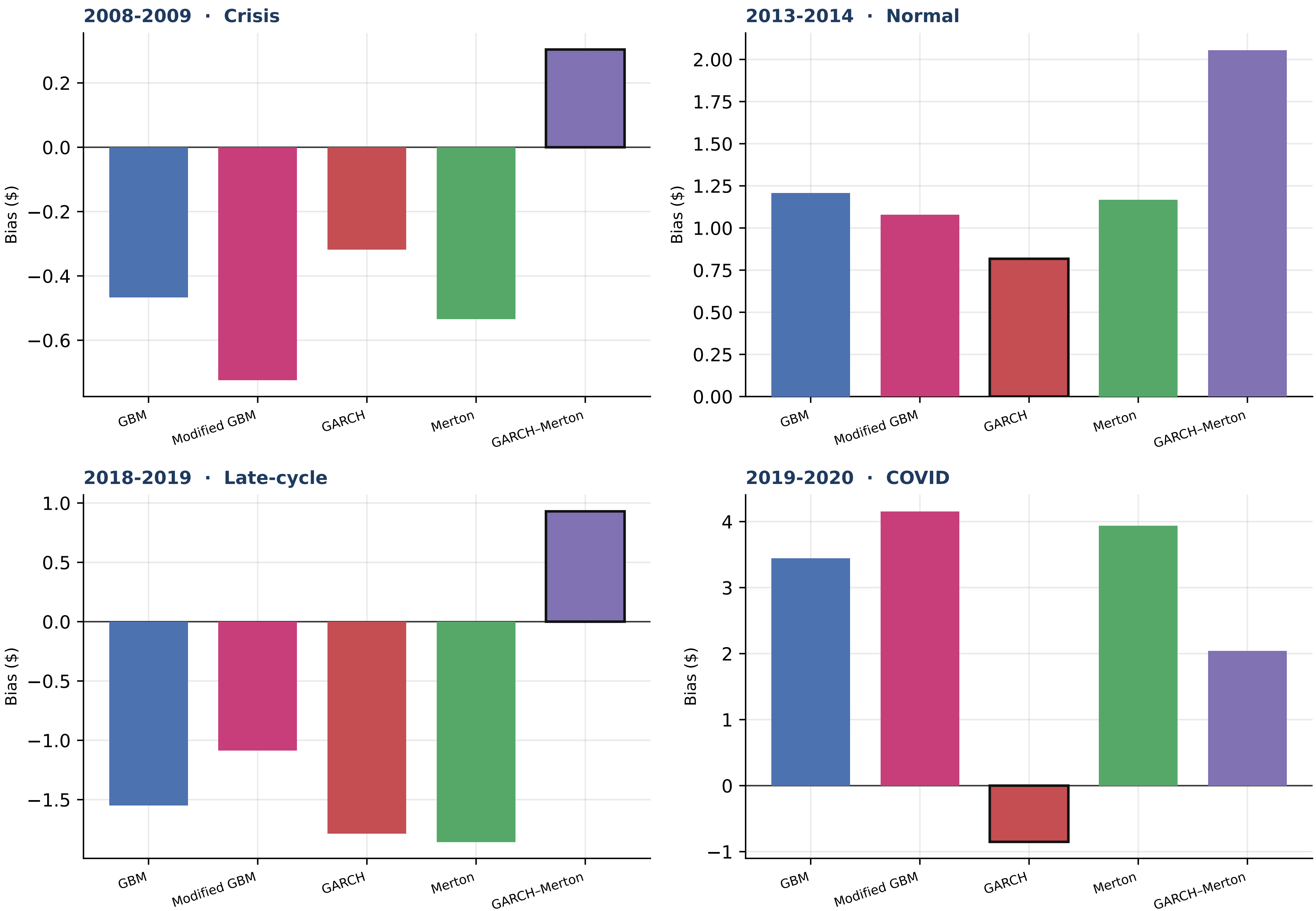
Outlined bar = lowest mean RMSE% for that company.

Page 11 · Figure 3 · Model rank by company × regime (1 = best RMSE%)



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.

Page 12 · Figure 4 · Bias (model − market, $) · mean of four companies



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, GARCH is the best overall model (mean RMSE% = 35.43; 3 of 16 cells).
- Second overall is Modified GBM (mean RMSE% = 37.46; 3 cells).
- The ranking is not stable. Best-in-cell models across the 16 tables: GARCH, GARCH–Merton, GBM, Merton, Modified GBM.
- SPY: lowest mean RMSE% is Merton (40.07); wins 2 of 4 regimes.
- AAPL: lowest mean RMSE% is GARCH (34.42); wins 1 of 4 regimes.
- MSFT: lowest mean RMSE% is Modified GBM (37.63); wins 2 of 4 regimes.
- AMZN: lowest mean RMSE% is GARCH (28.87); wins 1 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is GARCH (32.44).
- 2013-2014 (Normal): best mean RMSE% is GARCH (41.17).
- 2018-2019 (Late-cycle): best mean RMSE% is Modified GBM (20.89).
- 2019-2020 (COVID): best mean RMSE% is GARCH (46.57).
- No-jump models as a group have lower mean RMSE% (36.93) than the jump models in this report (41.98).
- For the overall winner (GARCH), crisis RMSE% is 32.44 vs 41.17 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year fixed calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1…MSFT-4 uses exactly the same option contracts for all five models in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias $, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).
- Early-exercise fractions are a model implication, not an accuracy score; American calls on these tenors often have modest early-exercise value.